

MindWealth — Portfolio Test Suite: Consolidated Hand-off

This single document folds the whole axiom-compliant test program — Phase A (engine + reconciliation), Phase B (follow-ups), and the Bucket-3 gaps — into one hand-off. Every number is produced by new, self-contained files; the delivered suite, the workbooks, and the core model were never edited. Universe throughout: dual-gated ($BT \geq 70\%$ AND $FWD \geq 60\%$), 179 assets, 4,461 trades. Costs = 15bps IBKR round trip; gross and net shown.

Executive summary — what we learned

- **The old numbers were flattering.** Averaging trade returns = rebalancing to equal weight every morning. On a true position-level, hold-original-weight book, chrono Sharpe is 0.62–0.67 vs the ~ 0.98 –1.26 the averaging construction printed for the same capped book (the workbook's 1.67 is monthly-sampled, a further flattering step). The whole CAGR/DD gap reconciles step by step (residual $\sim 0.4pp$).
- **Shorts earn their place — in bears.** Across 58,982 backtest trades to 1933, short combos are POSITIVE in every bear episode ($+0.98\%$ to $+2.95\%$) vs -0.91% in normal markets. The 2024–26 forward window (no bear) could never show this — this is the Axiom-6 evidence.
- **Eviction (1C) is real but the score is half-blind.** 1C adds Sharpe (admission + eviction both help) and keeps a younger book, but $\sim 40\%$ of held positions are underwater under every method — the score is built from backtest history and can't see a live position's age/P&L; That is Test 8's job.
- **The gate is stale.** Recomputing membership live fails the two approved FRACTAL shorts (Weekly 46.0%, Monthly 47.3%) against the 60% gate — the source of the 22-vs-19 discrepancy. (FRACTAL Daily Short also sits at 44.8%, though it was never in the approved list.)
- **The 1B quality threshold is NOT overfit** — it keeps 74–88% of its edge out-of-sample.
- **Blocked on others:** P3 replay + Test 6/8 final R:R + A1 real numbers + regime buckets all need Divyanshu; the score API is 401 (ledger-only scores used).

How every number is produced — the six axioms

- **1** NAV from position values (not averaged returns). **2** N self-financing slots, hold-original-weight to exit (no rebalance; leverage impossible). **3** pick on 2024, measure 2025–26. **4** gate recomputed at run time (57/63 hysteresis, stamped). **5** churn + dollar turnover + cost, gross & net. **6** protective mechanisms proven in bears, not averages.

Axiom 0 — Reconciliation waterfall (stat-test → workbook)

Note: Sharpe in this table uses $r_f=0$ uniformly across all steps so the deltas decompose cleanly; it differs slightly from the excess-return Sharpe quoted elsewhere. The DELTAS are the point of the table.

step	change	sharpe	cagr_%	maxDD_%	d_CAGR_pp	d_maxDD_pp
0	published: avg, full span, N=80	1.263	30.78	-29.78		
1	+ window 2024-2026 (avg, N=80)	0.743	16.84	-30.73	-13.94	-0.94
2	+ position-level hold-orig, N=80 (Ax1+2)	0.356	10.16	-13.73	-6.68	17.0
3	+ no-cap & rebalance (nav_engine=workbook B)	1.105	13.56	-13.26	3.4	0.47
4	+ month-end sampling (DD line)	1.591	13.56	-5.51	0.0	7.75
5	+ closed-only (workbook A)	2.4	26.47	-3.57	12.91	1.94

- Window is the biggest CAGR mover; construction+no-rebalance drops Sharpe and fixes DD (averaging overstated both); month-end sampling is the entire $-13\% \rightarrow -5.5\%$ DD gap. Lands on the workbook within $\sim 0.4pp$ — no hidden bug.

Part III — Repeat signals

No stacking cap (max depth 12); 18.8% of trades are repeats. **Repeats** $\approx$ **firsts** (win $\sim 76.6\%$ both) $\rightarrow$ keep as-is. Stacking inflates open-% for high-frequency combos.

Axiom 3 (1B walk-forward) + Axiom 4 (gate audit)

1B threshold: in-sample 2024 → out-of-sample 2025–26

threshold	IS_sharpe_2024	IS_n	OOS_sharpe_2025_26	OOS_n	retention_%
all	0.782	1372	0.136	2485	17.0
0	0.99	1003	0.602	1952	61.0
10	1.085	776	0.731	1506	67.0
20	1.201	548	1.06	1075	88.0
30	1.628	372	1.209	751	74.0
40	1.67	246	1.471	532	88.0
50	1.865	173	1.511	352	81.0
60	2.089	124	1.698	227	81.0

Not overfit — thresholds ≥ 20 keep 74–88% out-of-sample; the no-filter case collapses to 17%. (Correction: the grid is 0–60, not 0–40.)

Axiom 4 — combos below the live 60% gate (stale-list source)

Function	Interval	Signal	n	fwd_win_%
TRENDPULSE	Daily	Short	63	54.0
TRENDPULSE	Weekly	Short	34	52.9
FRACTAL TRACK	Quarterly	Short	27	48.1
FRACTAL TRACK	Monthly	Short	55	47.3
FRACTAL TRACK	Weekly	Short	161	46.0
FRACTAL TRACK	Daily	Short	672	44.8
FRACTAL TRACK	Daily	Long	20	30.0
DELTADRIFT	Monthly	Short	7	28.6
TRENDPULSE	Monthly	Short	7	14.3
TRENDPULSE	Quarterly	Long	4	0.0

P1 — Lemon test, F5 decomposition, open-book diagnostics

N	mode	sharpe	cagr_%	net_cagr_%	maxDD_%	calmar	churn_%	avg_age	pct_underwater
60	chrono	0.666	14.37	14.09	-13.11	1.1	0.0	415.0	39.5
60	rank	0.83	18.25	17.93	-15.78	1.16	0.0	426.0	36.0
60	evict	1.191	29.03	28.11	-17.06	1.7	46.4	268.0	41.0
120	chrono	0.623	13.32	13.0	-13.09	1.02	0.0	322.0	44.5
120	rank	0.885	18.19	17.85	-15.12	1.2	0.0	330.0	41.2
120	evict	1.061	24.0	23.34	-17.93	1.34	37.6	225.0	41.9

• F5: admission (rank–chrono) and eviction (evict–rank) both add Sharpe. Eviction keeps a younger book (415→268d). ~40% underwater under every method — the score-blindness hole (Test 8).

Open-% vs expected — combos flagged >1.5× (stacking-inflated, N=60)

Function	Interval	Signal	actual_open_%	expected_open_%	ratio
TRENDPULSE	Daily	long	10.01	0.85	11.77
DELTADRIFT	Daily	long	4.22	0.39	10.7
OSCILLATOR DELTA	Daily	long	1.85	0.59	3.13
DELTADRIFT	Daily	short	0.42	0.16	2.71
TRENDPULSE	Weekly	long	6.07	2.91	2.08
DELTADRIFT	Weekly	long	2.61	1.43	1.82
TRENDPULSE	Monthly	long	13.02	8.27	1.57

Trades per year (evaluated ≥30-clock window)

N	book	admitted	eval_years	trades_per_year
60	chrono	510	5.02	102.0
60	rank	501	5.02	100.0
60	evict	1581	5.02	315.0
120	chrono	1169	5.02	233.0
120	rank	1138	5.02	227.0
120	evict	2391	5.02	476.0

Test 8 — R:R risk flag on the 1C base

N	variant	sharpe	cagr_%	net_cagr_%	maxDD_%	calmar
60	baseline (1C)	1.191	29.03	28.11	-17.06	1.7
60	+admission gate	1.056	27.42	26.84	-18.74	1.46
60	+force-exit	1.053	24.61	23.41	-17.19	1.43
60	+both	0.752	19.03	18.25	-18.27	1.04
120	baseline (1C)	1.061	24.0	23.34	-17.93	1.34
120	+admission gate	1.053	23.68	23.28	-17.14	1.38
120	+force-exit	0.902	19.47	18.6	-16.6	1.17
120	+both	0.678	15.81	15.28	-17.02	0.93

• R:R is the Test-6 PLACEHOLDER (needs Divyanshu's compute_rr_to_nearest_support_stop). On the 2024–26 bull book it HURTS — admission blocks 53% of trades, CAGR/Sharpe fall; only N=120 force-exit trims MaxDD. Consistent with Axiom 6 (protection unproven without a bear). Motivates the real R:R + a bear test.

Axiom 6 — Behaviour in real bear markets (backtest ledger)

bucket	Signal	n	win_%	avg_ret_%
2000-02 dotcom	Long	268.0	63.8	85.99
2000-02 dotcom	Short	350.0	75.4	2.95
2007-09 GFC	Long	939.0	57.8	18.14
2007-09 GFC	Short	565.0	65.8	2.15
2018Q4	Long	1972.0	63.8	11.11
2018Q4	Short	454.0	67.0	1.42
2020 COVID	Long	1729.0	63.0	12.22
2020 COVID	Short	363.0	56.2	1.7
2022 bear	Long	6475.0	58.0	0.62
2022 bear	Short	1196.0	70.5	0.98
normal (non-bear)	Long	31831.0	72.5	8.6
normal (non-bear)	Short	12840.0	56.6	-0.91

• **Shorts are positive in EVERY bear** (+0.98% to +2.95%, win 56–75%) vs –0.91% in calm markets — the exact 'little in calm, a lot in stress' signature of a real hedge. 58,982 trades back to 1933. This is the single most important result the forward window could not produce.

Phase B — sizing, costs, robustness

A2 — Churn softener (eviction margin M)

N	margin_M	sharpe	net_cagr_%	maxDD_%	churn_%	dollar_turnover_%
60	0	1.191	28.11	-17.06	46.4	609.0
60	5	1.221	29.0	-18.68	42.4	586.0
60	10	1.204	28.91	-19.0	38.8	544.0
60	15	1.182	28.47	-18.34	35.3	502.0
60	20	1.169	28.06	-18.33	32.3	436.0
120	0	1.061	23.34	-17.93	37.6	441.0
120	5	1.02	23.06	-18.4	33.5	411.0
120	10	1.085	24.44	-18.67	29.9	401.0
120	15	1.037	22.49	-17.29	27.1	377.0
120	20	1.073	23.17	-17.35	24.5	369.0

Modest margin (M~5–10) cuts churn ~20–25% with no Sharpe loss — marginal evictions were pure cost.

US-listed-only rerun

scope	N	mode	trades	sharpe	cagr_%	net_cagr_%	maxDD_%	calmar	churn_%
ALL	60	chrono	4432	0.666	14.37	14.09	-13.11	1.1	0.0
ALL	60	rank	4432	0.83	18.25	17.93	-15.78	1.16	0.0
ALL	60	evict	4432	1.191	29.03	28.11	-17.06	1.7	46.4
ALL	120	chrono	4432	0.623	13.32	13.0	-13.09	1.02	0.0
ALL	120	rank	4432	0.885	18.19	17.85	-15.12	1.2	0.0
ALL	120	evict	4432	1.061	24.0	23.34	-17.93	1.34	37.6
US-only	60	chrono	2956	0.658	15.78	15.45	-15.2	1.04	0.0
US-only	60	rank	2956	0.84	21.59	21.22	-19.29	1.12	0.0
US-only	60	evict	2956	0.983	28.44	27.56	-21.19	1.34	43.6
US-only	120	chrono	2956	0.628	15.12	14.79	-16.09	0.94	0.0
US-only	120	rank	2956	0.79	18.54	18.18	-17.15	1.08	0.0
US-only	120	evict	2956	0.846	22.19	21.61	-20.1	1.1	33.5

Dropping non-US names (~33% of trades) lowers evict Sharpe (1.19→0.98) — international names add diversification.

Short signals vs shorting the S&P 500 (US-only)

Function	Interval	n	signal_avg_%	signal_win_%	shortSPX_avg_%	shortSPX_win_%	edge_avg_%	edge_win_%
DELTADRIFT	Daily	62.0	0.27	72.6	-0.05	51.6	0.32	64.5
DELTADRIFT	Weekly	20.0	1.2	75.0	0.5	45.0	0.7	65.0
FRACTAL TRACK	Monthly	37.0	0.08	51.4	-3.29	29.7	3.37	62.2
FRACTAL TRACK	Weekly	119.0	-1.79	45.4	-2.26	25.2	0.47	58.8
OSCILLATOR DELTA	Daily	169.0	-1.27	65.7	-0.59	44.4	-0.67	59.2
PULSEGAUGE	Weekly	113.0	-0.48	67.3	-0.4	43.4	-0.08	60.2
ALL		520.0	-0.84	61.5	-1.02	39.6	0.18	60.4

Gated shorts beat short-SPX by +0.18%/trade (edge in 60% of trades) — thin selection edge; both legs negative in the bull period.

Init-mode go-live test (avg over 6 dates)

N	mode	ret_6m_%	sharpe	maxDD_%
60	cold_ramp	9.24	1.81	-5.7
60	full_grab	12.48	1.81	-7.18
60	gated_seed	9.3	2.0	-5.9
120	cold_ramp	7.86	2.02	-4.65
120	full_grab	10.83	1.96	-6.35
120	gated_seed	8.07	1.99	-5.46

FULL GRAB = most return / deepest DD; GATED SEED = best risk-adjusted (recommend); COLD RAMP = safest / slowest.

P2 — Short stop-loss (per combo + portfolio)

Function	Interval	n	stop_%	orig_ER_%	stop_ER_%	triggered_%	ER_delta_%
FRACTAL TRACK	Weekly	161	7.7	-1.79	-1.22	29.2	0.57
DELTADRIFT	Weekly	37	10.0	0.59	0.68	21.6	0.09
DELTADRIFT	Daily	97	5.0	-0.35	-0.34	12.4	0.01
OSCILLATOR DELTA	Daily	237	10.0	-0.94	-0.95	29.5	-0.01
PULSEGAUGE	Weekly	142	9.9	-0.04	-0.31	11.3	-0.26
FRACTAL TRACK	Monthly	55	10.0	-0.49	-1.97	32.7	-1.48

N	book	sharpe	cagr_%	net_cagr_%	maxDD_%	calmar
60	baseline	1.191	29.03	28.11	-17.06	1.7
60	shorts-stopped	1.183	28.58	27.63	-17.2	1.66
120	baseline	1.061	24.0	23.34	-17.93	1.34
120	shorts-stopped	1.042	23.63	22.95	-17.93	1.32

Helps FRACTAL-Weekly / DELTADRIFT-Weekly; HURTS FRACTAL-Monthly (kills recovering winners). Portfolio-level neutral in 2024–26 — value is a bear-market question (Axiom 6).

A1 — Four-book sizing decomposition (PROXY inputs)

book	cagr_%	effect_pp
BASE (IBKR-connected)	13.41	
BASE+SSI	15.04	1.63
BASE+CONVICTION	15.56	2.15
ENHANCED	17.71	
= SSI effect		1.63
+ Conviction effect		2.15
+ interaction		0.52
= ENHANCED - BASE		4.3

Decomposition exact (residual 0). Inputs are proxies (VIX-adverse SSI, score-quartile conviction) — wire Divyanshu's real SSI-ceiling + conviction-tier feed for final numbers.

Realized (closed) vs MTM (open) drag, per combo

Function	Interval	Signal	closed_n	open_n	open_share_%	realized_ER_%	MTM_ER_%	open_ER_%	drag_pp
TRENDPULSE	Monthly	long	136	70	34.0	19.58	11.66	-3.72	-7.92
PULSEGAUGE	Weekly	long	178	91	33.8	14.03	8.04	-3.66	-5.99
DELTADRIFT	Monthly	long	5	1	16.7	16.77	12.48	-8.94	-4.29
TRENDPULSE	Weekly	long	300	108	26.5	8.96	4.73	-7.01	-4.23
BAND MATRIX	Daily	long	251	71	22.0	10.8	8.56	0.65	-2.24
DELTADRIFT	Weekly	long	124	10	7.5	7.2	5.43	-16.46	-1.77
TRENDPULSE	Daily	long	698	100	12.5	2.74	0.99	-11.23	-1.75
DELTADRIFT	Daily	long	538	50	8.5	2.67	1.68	-8.89	-0.99

The A(26%) vs B(14%) gap concentrates in underwater open longs (TRENDPULSE-Monthly –7.9pp).

Gap-close — final audit items

1B on position-level NAV (floating-N, Axiom 1) + IS/OOS

threshold	trades	avg_K	sharpe	cagr_%	net_cagr_%	maxDD_%	calmar	IS24_sharpe	OOS2526_sharpe	retention_%
all	942	68.0	0.704	14.41	14.13	-12.09	1.19	0.605	0.122	20.0
0	879	67.0	0.846	16.45	16.17	-12.88	1.28	0.926	0.563	61.0
10	825	62.0	0.873	17.82	17.52	-14.62	1.22	1.169	0.875	75.0
20	736	60.0	1.001	21.77	21.43	-16.72	1.3	0.991	0.92	93.0
30	678	57.0	1.094	26.15	25.76	-17.19	1.52	1.259	0.973	77.0
40	554	48.0	1.165	30.72	30.2	-17.5	1.76	0.576	1.213	211.0
50	421	37.0	1.231	35.62	35.11	-20.07	1.77	0.888	1.061	120.0
60	323	28.0	1.212	35.73	35.24	-20.19	1.77	0.967	1.02	105.0

Quality filtering survives the honest construction: Sharpe rises 0.70→1.23 with the threshold; the 2024 pick (thr 30) keeps 77% of its Sharpe in 2025–26 — not overfit.

A2 margin M — in-sample pick, out-of-sample confirm

N	margin_M	IS24_sharpe	OOS2526_sharpe	retention_%
60	0	1.191	1.192	100.0
60	5	1.198	1.206	101.0
60	10	1.085	1.13	104.0
60	15	0.987	1.042	106.0
60	20	1.052	1.03	98.0
120	0	0.997	0.948	95.0
120	5	1.014	0.894	88.0
120	10	0.952	0.892	94.0
120	15	1.021	0.847	83.0
120	20	1.082	0.791	73.0

N=60 pick M=5 retains 101% OOS; N=120 pick retains 73%. Both pass the 60% bar.

Gated-seed freshness sensitivity

hold_frac	price_drift_%	ret_6m_%	sharpe	maxDD_%
0.35	3.0	9.15	1.889	-5.57
0.35	5.0	9.41	1.985	-6.0
0.35	8.0	9.29	1.892	-6.34
0.5	3.0	9.49	1.919	-5.64
0.5	5.0	9.31	1.997	-5.9
0.5	8.0	8.87	1.827	-6.36
0.65	3.0	9.44	1.892	-5.68
0.65	5.0	9.47	1.973	-5.76
0.65	8.0	10.58	1.996	-6.1

Sharpe spread across the whole grid is 0.17 — the 50%-hold / 5%-drift starting numbers are robust.

Test 8 — removed set vs kept set (the P1 catch)

N	set	n	avg_age_d	age_vs_bthold	avg_unrealized_%	pct_underwater
60	KEPT by score (1C month-end holds)	3296	268.0	1.22	6.25	41.0
120	KEPT by score (1C month-end holds)	5485	225.0	1.16	5.0	41.9
-	REMOVED by R:R<1.0 (at forced exit)	1318	56.0	0.44	6.14	2.2

Win condition NOT met with the placeholder R:R: it removes YOUNG winners near target (56d, +6.1%, 2% underwater) — profit-taking — not the old underwater lemons the score keeps (268d, 41% underwater). This is the strongest evidence that Divyanshu's support-based R:R is required before Test 8 can do its job.

Same-asset cross-combo conflicts (Google-doc scenarios)

metric	n	win_%	avg_profit_%
trades opened INTO live opposite-direction position	892	68.2	0.44
trades opened with NO opposite-direction conflict	3569	78.7	5.1
same-direction, different-combo stacking entries	3174		
max concurrent trades on one asset (all combos)	15		

Material: 20% of trades open INTO a live opposite-direction position and average +0.44% vs +5.10% for clean entries (−4.65pp). Hold-both is costing; the conflict-handling rule deserves a design pass.

• **Daily snapshotter live** (`daily\_snapshot.py` → snapshots/YYYY-MM-DD.csv): 582 open positions, stamped rr_source=placeholder, 13% negative R:R shown never capped (corroborates the ~19% figure). History accrues from today — cannot be backfilled.

Deliverables & what's still blocked

- **Workbooks:** ..._GATED_FIXED_DAILYDD.xlsx (A & B) add true daily MaxDD, a Drawdown_Episodes sheet, filled CAGR, closed/MTM label, renamed 'Rolling CAGR'.
- **Reports:** MindWealth_Axiom_Report.pdf (Phase A), MindWealth_PhaseB_Report.pdf, this consolidated PDF, Portfolio.md + Portfolio_Axioms.md, P3_scoping.md.
- **Blocked on Divyanshu:** P3 point-in-time replay (core-code harness); Test 6/8 exact R:R (compute_rr_to_nearest_support_stop + no-clean-stop fallback); A1 real SSI/conviction feed; per-regime buckets (D1 daily series).
- **Blocked on infra:** composite-score API returns 401 — all scores here are ledger-only; live API scores need the auth fixed.
- **FEARFUL→BEARISH:** it is a data value + live match code (stest_common.py:406), not just report text — display renamed here; the match key must change with the data.